Sangola College, Sangola

Department of Computer Science & Applications

B.Sc (ECS) – III | Division B

ASSIGNMENT No. 01 – Data Mining & Preprocessing using WEKA

Subject:	Data Mining using WEKA	**Date:**	30 / 01 / 2025
Submission:	10 / 02 / 2025	**Tool:**	WEKA (Waikato Environment for Knowledge Analysis)

■ DATASET: Student.csv

Create a CSV file named **Student.csv** with the following data and load it in WEKA:

Sid	Sname	Course	Gender	Age	Attendance
101	Karan	BCA	Male	20	high
102	Anjali	ECS	Female	18	medium
103	Rohit	MCA	Male	21	low
104	Kajol	MSC	Female	22	low
105	Shruti	BA	Female	18	medium
106	Satish	BCA	Male	19	high
107	Laxman	BA	?	?	?
108	Sarang	?	Male	?	?
109	Kapil	BCOM	?	?	low
110	Harsha	ECS	Female	?	?

***Note:** Cells marked with **?** represent missing values. In WEKA, missing values are represented by **'?'** in the dataset.*

■ SOLUTIONS TO ALL 20 QUESTIONS

Q1. Load Dataset

To load the Student.csv dataset in WEKA:

- ■ Open WEKA GUI Chooser → Click on **Explorer**
- ■ In the **Preprocess** tab → Click **'Open file...'**
- ■ Browse and select **Student.csv** → Click Open
- ■ The dataset will load showing 10 instances and 6 attributes: Sid, Sname, Course, Gender, Age, Attendance

■ *Note: WEKA auto-detects attribute types: numeric for Sid & Age; nominal for others.*

Q2. Add new attribute 'Grade' at index 5 with Nominal data type having {Dist., First}

Use the **AddValues** or **Add** filter under weka.filters.unsupervised.attribute:

- ■ Preprocess tab → Click **'Choose'** under Filters
- ■ Navigate to: **unsupervised** → **attribute** → **Add**
- ■ Set parameters: **attributeIndex = 5**, **attributeName = Grade**, **attributeType = NOM**
- ■ Set **nominalLabels = Dist.,First**
- ■ Click **Apply** → New attribute 'Grade' appears at position 5

■ *Note: Attribute index is 1-based in WEKA. Index 5 places 'Grade' between Age and Attendance.*

Q3. List the attribute names

After loading the dataset, attribute names can be viewed in WEKA:

- ■ In the Preprocess tab, the **Attributes panel** (left side) lists all attributes
- ■ Click on each attribute to see its details in the right panel

#	Attribute Name	Type	Description
1	Sid	Numeric	Student ID
2	Sname	Nominal/String	Student Name
3	Course	Nominal	Course enrolled
4	Gender	Nominal	Male / Female
5	Age	Numeric	Student Age
6	Attendance	Nominal	high / medium / low

Q4. Find an attribute which has 30% missing values in the dataset

To find missing value percentages in WEKA:

- In Preprocess tab, click on each attribute in the Attributes list
- Check the **'Missing'** count shown in the attribute statistics panel

Attribute	Missing Count	Missing % (out of 10)	Status
Sid	0	0%	Complete
Sname	0	0%	Complete
Course	1 (Sarang)	10%	OK
Gender	1 (Laxman)	10%	OK
Age	3 (Laxman, Sarang, Harsha)	30%	**30% MISSING**
Attendance	3 (Laxman, Sarang, Harsha)	30%	**30% MISSING**

Answer: Age and **Attendance** attributes each have **30% missing values** (3 out of 10 records).

Q5. Remove any attributes if the missing values are more than 50%

Check: No attribute has more than 50% missing values in this dataset.

However, if there were such attributes, the method to remove is:

- Filter → **unsupervised** → **attribute** → **RemoveUseless**
- Or manually: **unsupervised** → **attribute** → **Remove** → specify attribute index
- Set the attribute index to remove → Click **Apply**

Note: In this dataset, no attribute exceeds 50% missing, so NO attribute is removed. Age and Attendance have only 30% missing values.

Q6. Convert 'Sid' attribute numeric value to nominal

Use the **NumericToNominal** filter to convert Sid from numeric to nominal:

- Preprocess → Filter → Choose → **unsupervised** → **attribute** → **NumericToNominal**
- Click on the filter name to open its configuration
- Set **attributeIndices = 1** (Sid is at index 1)
- Click **OK** → then click **Apply**
- Verify: Sid attribute type changes from **Numeric** to **Nominal**

Note: This is useful because Sid is an identifier, not a measurable quantity. Treating it as nominal prevents WEKA from performing arithmetic on it.

Q7. Replace missing data if there is any

Use the **ReplaceMissingValues** filter:

- Preprocess → Filter → Choose → **unsupervised** → **attribute** → **ReplaceMissingValues**
- This filter replaces missing values with the **mean** (for numeric attributes) and **mode/most frequent value** (for nominal attributes)
- Click **Apply** → All '?' values are replaced

Attribute	Type	Missing Values	Replaced With

Course	Nominal	? (Sarang)	Mode = BCA or ECS
Gender	Nominal	? (Laxman)	Mode = Male
Age	Numeric	? (Laxman, Sarang, Harsha)	Mean ≈ 19.67
Attendance	Nominal	? (Laxman, Sarang, Harsha)	Mode = medium

Q8. 'Age' attribute into three bins (intervals)

Use the **Discretize** filter to bin the Age attribute into 3 intervals:

- Filter → Choose → **unsupervised** → **attribute** → **Discretize**
- Set **attributeIndices = 5** (Age is at index 5)
- Set **bins = 3**
- Set **useEqualFrequency = False** (use equal width bins)
- Click OK → Apply

Age range: 18–22. Three equal-width bins:

Bin	Range	Label (WEKA)	Students
Bin 1	18 – 19.33	'(-inf-19.33]'	Anjali(18), Shruti(18), Satish(19)
Bin 2	19.33 – 20.67	'(19.33-20.67]'	Karan(20)
Bin 3	20.67 – 22	'(20.67-inf)'	Rohit(21), Kajol(22)

Q9. Copy attribute Age at 3rd position

Use the **Copy** filter to duplicate an attribute and place it at a specific index:

- Filter → Choose → **unsupervised** → **attribute** → **Copy**
- Set **attributeIndices = 5** (Age attribute index)
- Click Apply → This adds a copy of Age
- Then use **Reorder** filter to place it at position 3
- Filter → **unsupervised** → **attribute** → **Reorder**
- Set indices to: **1,2,7,3,4,5,6** (placing copied Age at 3rd position)

■ *Note: The exact index of the copied attribute depends on current attribute count. Adjust the Reorder indices accordingly.*

Q10. Check whether there is an outlier or extreme value in the dataset

Use the **InterquartileRange** filter to detect outliers:

- Filter → Choose → **unsupervised** → **attribute** → **InterquartileRange**
- Set **attributeIndices = 5** (Age attribute) or 'first-last' for all numeric
- Set **outlierFactor = 3**, **extremeValuesFactor = 6**
- Click Apply → WEKA adds two new binary attributes: 'Outlier' and 'ExtremeValue'

Manual Check using IQR method for Age:

Statistic	Value
Sorted Ages (known)	18, 18, 19, 20, 21, 22
Q1 (25th percentile)	18.5
Q3 (75th percentile)	21
IQR = Q3 - Q1	2.5
Lower Bound (Q1 - 1.5*IQR)	18.5 - 3.75 = 14.75
Upper Bound (Q3 + 1.5*IQR)	21 + 3.75 = 24.75
Outlier in Dataset?	No regular outliers in valid ages
Extreme Values (e.g., 200, 220)	**YES – Rows 107 (200) and 109 (220) are EXTREME values**

Q11. Remove outliers and extreme values whose values are 0

After applying InterquartileRange filter (Q10), use **RemoveWithValues** filter:

- Filter → Choose → **unsupervised** → **instance** → **RemoveWithValues**
- Set **attributeIndex** to the 'Outlier' attribute column added in Q10
- Set **nominalIndices = 1** (value '1' = outlier flag)
- Click Apply → Removes instances flagged as outliers/extreme values

Alternative method: Use **unsupervised** → **instance** → **RemoveRange** to manually remove rows 107 and 109 which have age values 200 and 220.

■ *Note: 'Whose values are 0' likely refers to removing the extreme/outlier flag where the outlier indicator is set. Rows with age 200 and 220 are removed.*

Q12. Rescale 'Age' attribute

Use the **Normalize** or **Standardize** filter to rescale Age:

Method 1: Normalize (Min-Max Scaling to [0,1])

- Filter → **unsupervised** → **attribute** → **Normalize**
- Set scale = 1.0, translation = 0.0 (default for 0 to 1 range)
- Set attributeIndices = 5 (Age)
- Click Apply → Age values rescaled between 0 and 1

Method 2: Standardize (Z-score normalization)

- Filter → **unsupervised** → **attribute** → **Standardize**
- Converts Age to mean=0, std=1

Original Age	Normalized (0-1)	Formula: (x-min)/(max-min)
18	0.00	(18-18)/(22-18) = 0.00
19	0.25	(19-18)/(22-18) = 0.25
20	0.50	(20-18)/(22-18) = 0.50
21	0.75	(21-18)/(22-18) = 0.75
22	1.00	(22-18)/(22-18) = 1.00

Q13. Reorder the data

Use the **Reorder** filter to rearrange attribute order:

- Filter → Choose → **unsupervised** → **attribute** → **Reorder**
- In the 'attributeIndices' field, specify the new order as comma-separated indices
- Example – reorder to: Sname, Gender, Age, Course, Attendance, Sid
- Enter: **2,4,5,3,6,1** → Click OK → Apply

To sort instances (rows) by attribute value, use **Sort** filter:

- Filter → **unsupervised** → **instance** → **Sort**
- Set attributeIndex = 5 (Age) to sort rows by Age in ascending order

Q14. Removing duplicates

Use the **RemoveDuplicates** filter to eliminate duplicate instances:

- Filter → Choose → **unsupervised** → **instance** → **RemoveDuplicates**
- No additional configuration needed → Click Apply
- WEKA will scan all instances and remove any exact duplicate rows

■ *Note: In the Student.csv dataset, there are no exact duplicate rows. This step serves as a verification/safeguard during preprocessing.*

Q15. Apply Attribute Selection

Attribute selection removes irrelevant/redundant attributes to improve model performance:

- Go to **Select Attributes** tab in WEKA Explorer
- Attribute Evaluator: Choose **CfsSubsetEval** (Correlation-based Feature Selection)

- Search Method: Choose **BestFirst** or **GreedyStepwise**
- Click **Start** → WEKA evaluates and ranks attributes

Alternative using Filter-based method:

- Preprocess → Filter → **supervised** → **attribute** → **AttributeSelection**
- Set evaluator to **InfoGainAttributeEval** and ranker to **Ranker**
- Apply → Only top-ranked attributes are retained

■ *Note: Sid (student ID) and Sname (name) would likely be removed as they carry no predictive information.*

Q16. Replace the missing value with constant value — Change numeric value with 1 and nominal value with N/A

Use the **ReplaceWithMissingValue** combined with **ReplaceMissingValues**:

For replacing with a **specific constant**, use:

- Filter → **unsupervised** → **attribute** → **ReplaceWithMissingValue** (first mark them)
- Then use a custom approach: Filter → **unsupervised** → **attribute** → **StringToNominal** and set default values

Practical Method in WEKA:

- For Numeric missing values → Use **MathExpression** filter to assign value = 1
- For Nominal missing values → Manually edit the CSV: replace all '?' with 'N/A'
- Reload the modified CSV in WEKA

Attribute	Type	Missing Value	Replaced With
Age	Numeric	?	1
Course	Nominal	?	N/A
Gender	Nominal	?	N/A
Attendance	Nominal	?	N/A

Q17. To retrieve all instances (rows) from dataset whose age is above 20

Use the **RemoveWithValues** filter to keep only rows where Age > 20:

- Filter → **unsupervised** → **instance** → **RemoveWithValues**
- Set **attributeIndex = 5** (Age)
- Set **splitPoint = 20**
- Set **invertSelection = True** (to KEEP values above 20, not remove them)
- Click Apply

Instances with Age above 20:

Sid	Sname	Course	Gender	Age	Attendance
103	Rohit	MCA	Male	21	low
104	Kajol	MSC	Female	22	low

Result: 2 instances retrieved (Rohit - Age 21, Kajol - Age 22)

Q18. Remove 20% instances of dataset using Remove Percentage filter

Use the **RemovePercentage** filter:

- Filter → Choose → **unsupervised** → **instance** → **RemovePercentage**
- Set **percentage = 20**
- Set **invertSelection = False** (removes 20%, keeps 80%)
- Click OK → Apply

Calculation:

- Total instances = 10
- 20% of 10 = 2 instances will be removed
- Remaining instances = 8 (the first 8 rows are kept by default)

■ *Note: WEKA removes the LAST 20% of instances by default. To remove random instances, shuffle data first using Randomize filter.*

Q19. Produce 15% random resample of dataset

Use the **Resample** filter for random resampling:

- ■ Filter → Choose → **unsupervised** → **instance** → **Resample**
- ■ Set **sampleSizePercent = 15**
- ■ Set **randomSeed = 1** (for reproducibility)
- ■ Set **noReplacement = True** (sample without replacement)
- ■ Click OK → Apply

Calculation:

- ■ Total instances = 10
- ■ 15% of 10 = 1.5 ≈ 2 instances will be in the resample
- ■ Result: A new dataset with approximately 1–2 randomly selected instances

■ *Note: With replacement (noReplacement=False), the same instance can appear multiple times. Without replacement, each instance appears at most once.*

Q20. To change name of attribute Sname to Stud_Name

Use the **RenameAttribute** filter to rename an attribute:

- ■ Filter → Choose → **unsupervised** → **attribute** → **RenameAttribute**
- ■ Set **attributeIndex = 2** (Sname is the 2nd attribute)
- ■ Set **newName = Stud_Name**
- ■ Click OK → Apply
- ■ Verify: In the Attributes panel, 'Sname' is now renamed to 'Stud_Name'

■ *Note: An alternative method: Go to Edit → Rename Attribute directly in the WEKA Explorer. Or use the ARFF editor to manually edit the @attribute declaration.*

■ QUICK REFERENCE SUMMARY

Q#	Task	WEKA Filter / Method
1	Load Dataset	Explorer → Preprocess → Open file
2	Add Grade attribute	unsupervised → attribute → Add
3	List attributes	Preprocess → Attributes panel
4	Find 30% missing	Click attribute → check Missing count
5	Remove >50% missing	unsupervised → attribute → Remove
6	Sid numeric to nominal	unsupervised → attribute → NumericToNominal
7	Replace missing values	unsupervised → attribute → ReplaceMissingValues
8	Age into 3 bins	unsupervised → attribute → Discretize (bins=3)
9	Copy Age at 3rd position	unsupervised → attribute → Copy + Reorder
10	Check outliers	unsupervised → attribute → InterquartileRange
11	Remove outliers	unsupervised → instance → RemoveWithValues
12	Rescale Age	unsupervised → attribute → Normalize/Standardize
13	Reorder data	unsupervised → attribute → Reorder
14	Remove duplicates	unsupervised → instance → RemoveDuplicates
15	Attribute selection	Select Attributes tab → CfsSubsetEval
16	Replace with constant	Edit CSV (N/A / 1) or MathExpression filter
17	Age > 20 instances	unsupervised → instance → RemoveWithValues (invert)

18	Remove 20% instances	unsupervised → instance → RemovePercentage (20%)
19	15% random resample	unsupervised → instance → Resample (15%)
20	Rename Sname attribute	unsupervised → attribute → RenameAttribute